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4 January 1966

MILITARY SPECIFICATION
PUMP , FUEL

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope. This specification covers one type of swashplate piston-type fuel pump referenced herein as the "pump." The pump has a range of variable discharge pressures controlled by a bias pressure.

2. Applicable DOCUMENTS

2.1 Applicable documents. The following documents of the issue in effect on date of invitation for bids or req uest for proposal form a part of this specification to t-he extent specified heretin.

SPECIFICATIONS

Military	
MIL-P-116	Preservation. Methods of
MIL-G-4343	Grease, Pneumatic System
MIL-L-15016	Lubricating Oil
MIL-R-18866	Fittings, Flared Tube, 37-Degree, Steel
WS-2106	Fuel, Liquid Monopropellant, Otto Type II

STANDARDS

Military	
MIL-STD-105	Sampling Procedures and Tables for Inspection by attributes
MIL-STD-12°	Marking for Shipment and Storage
MS-16626	Ring, Retaining, External Inverted "Tapered Section Type".
MS-20995	Wire, Lock
MS-29561	Ricking, Preformed, C Ring Synthetic Lubricant Resistant
MS-35670	Fittings, Lubricant Pressure Relief
MS-51964	Setscrew-Hex Socket, Cup Point, Alloy Steel, Cad Plate UNF-3A

DRAWINGS

Bureau of Naval Weapons -	
DL-1568925	Pump Assembly, Fuel
2526741	Nut, Adjusting Screw

FSC 1355

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CD-15 56991	Spring, Ball Guide
CD-1 557047	Spring, Piston
CD-1557048	Bearing, Needle
CD-1557052	Packing, Preformed
CD-1557063	Valve Assembly
CE-1557065	Shaft, Pump
CD-1568879	Body, Fuel Pump
CD-1568892	Block, Piston
CD-1568924	Head Pump
CD-1568925	Pump Assembly, Fuel
CD-2526584	Seal, Lower
CD-2526593	Piston
CD-2526597	Cam
CD-2526611	Screw, Cap Sot. Hd. Hex
CD-2526612	Retainer Seal
CD-2526613	Plate, Thrust
CD-2526614	Bearing, Needle
CD-2526616	bearing, Thrust
CD-2526617	Washer, Thrust
CD-2526624	Seal, Plain Encased
CD-2526718	Packing, Preformed
CD-2526738	Pin, Valve
CD-2526739	Guide, Ball
CD-2526740	Screw, Calve, Adj.
CD-2526741	Nut, Adj. Screw
CD-2526742	Ball
CD-2526834	Seal, Upper
CD-2571159	Key, Woodruff
CD-2526724	Head Assembly

PUBLICATIONS

Bureau of Naval Weapons

OP-3368	Safety and Handling Instructions. Otto Fuel II

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procuring functions should be obtained from the procuring activity or as directed by the contracting officer.)

Nat ionalAircraft Standards

NAS-617	Packing Preformed
NAS-620	Washer, Flat
NAS-1351	Screw, Cap, Soc Hd. Hex

(Copies of National Aircraft Standards maybe obtained from the National Standards Association, Inc. 1321 Fourteenth Street, N. W. Washington, D.C. 20005.)

Aerospace Recommended Practice

ARP-568-008	Packing Preformed

(Copies of National Aircraft Standards maybe obtained from the Society Automotive Engineers, 485 Lexington Avenue, New Yori, New York 10017.)

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3. REQUIREMENTS

3.1 Classification of requirements. The requirements for the pump are classified as follows:

REQUIREMENT	PARAGRAPH
General Requirements	3.2
Per formance Requirements and Product Characteristics	3.3
Environmental Requirements	3.4
Operational Service Life	3.5

3.2 General requirements.

3.2.1 Data requirements. No data is required by this specification or approval documents referenced in Section 2 unless specified in contract or purchase order. See 6.2.

3.2.2 Preproduction sample. Unless otherwise specified in the contract or purchase order, the preproduction sample of the pump shall be manufactured using the methods and procedures proposed for production. The sample will be tested as specified in Section 4 herein and is for the purpose of determining that prior to starting production, the contractor's production methods are capable of yielding items that comply with the technical requirements of the concract.

3.2.3 Materials. Unless otherwise specified in the contract or purchase order, all materials and treatments of materials and parts used in the manufacture of the pump shall be in accordance with the applicable documents listed on BuWeps DL 1568925.

3.2.4 Definitions. Definitions of terms appear in Section 6.

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3.3 Performance requirements and Product characteristics. Unless specified otherwise, the pumps shall satisfy the following performance requirements when tested as specified in Section 4.

3.3.1 Liquid The pump shall meet the performance requirements of 3.3.2 using Otto fuel type II, WS-2106, or a 5 percent solution of soluble oil in tap water.

3.3.2 Flow rates and pressure values. The pump when driven at 3880 ± 10 revolutions per minute (rpm) and limited to flow rates defined in Table I shall be capable of producing the respective output pressures listed in Table I when operated at the corresponding bias pressure.

TABLE I

OPERATING CHARACTERISTICS

Nominal Flow Rate gpm	Bias Pressure Psig	Output Pressure Psig
1.25	0 (Atmospheric)	2345 ± 100
1.28	85 ± 10	2475 ± 75
1.33	225 ± 10	2675 ± 75
1.40	445 ± 10	2925 ± 75
1.45	665 ± 10	3155 ± 100
1.54 Min.	1100 ± 10	3575 Min.

3.3.3 Leakage . The pump shall show no sign of external leakage of working fluid while operating.

3.4 Environmental requirements. Unless otherwise specified, the pump shall be capable of conforming to the performance requirements and product characteristics of the specification after being exposed to the environmental conditions specified in this section. For definitions of ready, operating, and non-operating see 6.3.1, 6.3.2, and 6.3.3.

3.4.1 Vibration, ready condition. The pump, mounted as shown in Figure 1, shall be capable of meeting the performance requirements after being subjected, in the resay condition, to the vibrations defined in Table II. The pump shall be vibrated for 180 minutes each in its longitudinal axis and in two mutually perpendicular transverse axes at any temperature within the range of -45° F to +140° F. Vibration is defined in 6.3.4.

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TABLE II

VIBRATION SCHEDULE

Double Amplitude (in. ± 10%) Minimum	Acceleration (g's ± 10%) Minimum	Frequency (cps)
0.2	---	5 - 22
---	3.0	22 - 40
0.036	---	40 - 74
---	10	74 - 500

3.4.2 Acceleration, ready condition. The pump, mounted as shown in Figure 1, shall be capable of meeting the performance requirements after being subjected, in the ready condition, to the accelerations defined in Table III for a duration of 1 minute for each magnitude. Acceleration, is defined in 6.3.5.

TABLE XII

ACCELLARATION SCHEDULE

Magnitude (g's ± 10%)	Acceleration Direction (See Figure 1)
25	Longitudinal axis, Fwd
35	Longitudinal axis, Aft

3.4.3 Shock, ready condition. The pump, mounted as shown in Figure 1, shall be capable of meeting the performance requirements after being subjected, in the ready condition, to the shock loads defined in Table IV. Shock is defined in 6.3.6.

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TABLE IV

SHOCK SCHEDULE

Test Sequence	Peak Magnitude (g's ± 10%)	Duration (ms)	Impact Axis	Direction Of Deceleration Force	Number Of Impacts
1	300	0.5 to 1.5	Longitudinal	Aft	3
2	85	0.5 to 1.5	Transverse A	Optional	3
3	85	0.5 to 1.5	Transverse B	Optional	3
4	45	18.0 to 20.0	Longitudinal	Fwd	3
5	60	8.0 to 10.0	Longitudinal	Aft	3
6	60	8.0 to 10.0	Longitudinal	Fwd	3
7	60	8.0 to 10.0	Transverse A	Optional	3
8	60	8.0 to 10.0	Transverse B	Optional	3

3.4.4 Temperature and humidity.

3.4.4.1 Non-operating.

a. Temperature. The pump shall be capable of meeting the performance requirements subsequent to withstanding, in the non-operating condition, environmental temperatures between -45° F and +140° F. Temperature is defined in 6.3.7.

b. Humidity. The pump shall be capable of meeting the performance requirements subsequent to withstanding, in the non-operating condition, an environment of at least 95 percent relative humidity at +110° F for a minimum period of 8 hours. Humidity is defined in 6.2.8.

3.4.4.2 Ready, operating. The pump. immediately after being conditioned for 1 hour at any temperature between 0° F and +140° F shall be capable of meeting the performance requirements at any enviromental temperature between 0° F and +140° F.

3.5 Operational service life. The pump shall have a minimum service life of 70 minutes which shall be divided into 10 cycles of 7 minutes each.

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3.6 Workmanship. The workmanship and finish shall be of sufficiently high grade to insure satisfactory operation consistent with the requirements of this specification. The assembly shall be thoroughly clean and free of excess materials, chips, and loose or spattered foreign materials. The standards of workmanship exhibited in any approved pre-production sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PREVISIONS

4.1 Responsibility. for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspection are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection of the pump shall be classified as follows:

a. Preproduction inspection (Subsection 4.4)
b. Acceptance inspection (Subsection 4.5)
c. Periodic inspection (Subsection 4.6)
d. Examination of first echelon repair parts (Subsection. 4.7)

4.3 Sampling.

4.3.1 Preproduction inspection sample. A preproduction sample of 3 pumps manufactured in accordance with 3.2 shall be subjected to the preproduction tests listed in Table V and detailed in 4.4 through 4.4.4.10 at an activity designated by the procuring activity. Acceptance of the preproduction sample shall be based on no defects in the sample. See 4.4.6. Further production of the pump by the supplier, prior to approval of the preproduction sample shall be at the supplier's risk.

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4.3.2 Periodic production sample. Periodic production samples shall be selected from units that have satisfactorily passed the acceptance inspections of 4.5. These samples shall be subjected to the tests of 4.6 at an activity designated by the procuring activity. Periodic production sampling shall be as follows:

Step 1. Upon initiation of production randomly select 1 unit from each group of 30 units. Upon satisfactory compliance of samples selected from 3 consecutive groups, shift to Step 2.

Step 2. Randomly select 1 unit from each group of 70 units. Continue with Step 2 for the balance of the contract or until a defective unit is found.

When a defective unit is found in any group, acceptance of all units on hand shall be withheld and the contractor shall:

a. Determine the cause of the defect.

b. Take the necessary actions to correct or eliminate the defect from all units on hand and, if so directed by the procuring activity, from those units previously shipped.

c. Submit to the procuring activity for approval, the proposed corrective action to prevent recurrence of the defect.

d. Submit to the procuring activity for approval the proposed retest program to verify that the corrective action eliminated the defect.

Return to Step 1 and proceed as indicated when inspection is resumed.

4.3.3 Examination of first echelon repair parts. The sampling plans and procedures set forth in MIL-STD-105 shall be used in the determination of acceptability of first echelon repair parts for the pump as specified in 4.7 through 4.7.4.

4.4 Preproduction inspection.

4.4.1 Tests and examinations. The preproduction sample of 3 pumps shall be subjected to the operational test defined in 4.5.4.1 and subsequently shall be inspected in accordance with Table V, with 1 pump being tested in accordance with each of the 3 test sequences, A through C.

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TABLE V

PREPRUDUCTION TEST SEQUENCE

Test	Paragraph Reference	Test Sequence A	B	C
Operational Test	4.4.4.1	1	1	1
Vibration, Ready Condition	4.4.4.2	2	2	
Acceleration, Ready Condition	4.4.4.3	3		
Shock, Ready Condition	4.4.4.4	4		
Operational Test	4.4.4.1	5	3	
Temperature and Humidity	4.4.4.5	6	4	
Low Temperature, Operational Test	4.4.4.6	7		
High Temperature, Operational Test	4.4.4.7		5	
Vibration, Packed and Packaged	4.4.4.8			2
Shock, Packed and Packaged	4.4.4.9			3
Operational Service Life	4.4.4.10	8	6	4
Pump "A" Seven operational tests "B" Seven operational tests "C" Nine operational tests				

4.4.2 Test conditions. Unless otherwise specified, the pumps shall be subjected to the tests in 4.4.1 under the following conditions:

a.	Temperature:	Room ambient +32° F to +98° F
b.	Altitude:	Normal ground atmospheric pressure
c.	Vibration:	None
d.	Humidity:	Room ambient

4.4.3 Test equipment and set-up. The following items of equipment are required for the preproduction tests.

4.4.3.1 Test stand. The instrumentation required to obtain and record the pump performance data shall have a total range commensurate with the testing range and accuracies as indicated, An oscillographic recorder and associated transducer shall be used in addition to the visual readout equipment for the functions so indicated. The readout error of the oscillographic recording system shall be ±2 percent, maximum. The pre-production test equipment snail be assembled in accordance with Figure 2. When performing operational tests using Otto fuel, all components containing Otto fuel shall be enclosed by a barrier, thus providing protection for personnel.

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a. Fuel delivery system. The fuel delivery system shall be capable of delivering Otto fuel to the inlet port of the pump at a rate of 2 gallons per minute (gpm), at a pressure of 7 ±3 pounds per square inch, gage (psig), and at ambient temperatures of +140° F ± 5° F and 0° F ± 5° F during operation. The fuel may be conditioned to the low temperature, 0° F ± 5" F, by use of the inlet heat exchanger or by cooling the bulk fuel stored in the fuel supply tank. For the high temperature requirement, +140° F ± 5' F, the fuel must be conditioned by the inlet heat exchanger. This heat exchanger should be located as close as possible to the inlet port of the fuel pump. The pressure and temperature shall be measured at the inlet port to the pump. The relief valve "A", and the solenoid valve shall be coupled to the fuel supply tank with a minimum of connections and tubing.

Gage A Range: 0 to 25 psig
Accuracy: ± 5% full scale

Relief Valve A Cracking Pressure: 25 ± 3 psig

Temperature sensor, inlet
Range -10°F to +180° F nominal

Pressure sensor, inlet
Range 0 to 1.5 psig nominal

b. Drive unit. 5 horsepower synchronous electric, nominal

c. Bias pressure control.

Regulated inert gas supply; Output range, 0 to 1250 psig

Gage:
Range: 0 to 1250 psig nominal
Accuracy: Within ± 1% full scale
Size: 6 inch minimum

Pressure sensor:
Range: 0 to 1250 psig nominal

d. Fuel discharge system. The fuel discharge system shall contain the following components:

Pressure sensor, outlet:
Range: 0 to 5000 psig

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Relief valve B: Cracking pressure 4500 ± 100 psig
Orifice, Flow limiting: The fixed orifice, when connected in the test set-up, Figure 2, shall produce a pressure at a flow rate as defined by:

$P = KQ^2 \quad P > 2500$ psig
P = pressure - psig
Q = flow, gallons per minute of fuel
K = 1500 ± 94

e. Discharge storage tank. Discharge Otto fuel shall be collected in a storage tank. The Otto fuel shall be disposed of in accordance with NavWeps OP 3368.

THE OTTO FUEL SHALL NOT BE USED AGAIN FOR TESTING PUMPS OR FOR ANY OTHER OPERATIONAL TESTS

4.4,3.2 Vibration equipment with temperature chamber. The vibration equipment with temperature chamber shall have the following characteristics:

a. Frequency range, logarithmic sweep: 5 to 500 cycles per second (cps)
b. Test specimen: 8 pounds
c. Acceleration and displacement (ready condition):

5 to 22 cps	0.2 inch
22 to 40 cps	3.0 gravitational units (g's)
40 to 74 cps	0.036 inch
74 to 500 cps	10 g's

d. Acceleration (packed and packaged):

5 to 60 cps	l g

e. Automatic cycling time range: 0 to 15 minutes (optional)
f. Temperature chamber range: -45° F to +140° F accuracy to ± 5° F
g. Interior size: 7-1/2 inches x 6 inches x 8 inches minimum

4.4.3.3 Acceleration equipment. The acceleration equipment shall have the following characteristics: 35 g's with an 8 pound test specimen.

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4 .4.3.4 Shock equipment . The shock equipment shall be capable of accepting an 8 pound test specimen and shall provide ranges of:

a. 300 g's for 0.5 milliseconds (ins) to 1.5 ms
b. 60 g's for 8.0 ms to 10.0 ms
c. 45 g's for 18.0 ma to 20.0 ms

4.4.3.5 Temperature and humidity conditioning equipment. One or more equipments may be utilized to provide the following characteristics:

a.	Temperature:	-45° F to +140° F; with an accuracy to ± 5° F
b.	Relative humidity:	95 percent minimum at +110° F; with an accuracy to ± 5° F
c.	Interior size:	7-1/2 inches x 6 inches x 8 Inches minimum

This equipment may be used for both non-operating and ready/operating temperature (and humidity) tests.

4,4.4 Test procedures .

4.4.4.1 Operational test. The preproduction pumps shall be mounted in the pump test stand and connected to the auxiliary equipment and instruments in accordance with Figure 2. Twenty cubic centimeters (2/3 oz.) of lubrication oil, see 6.7, shall be used in the pump housing. The pump and all items containing Otto fuel shall be enclosed by a suitable barrier. The controls and instrumentation readout indicators shall be remotely located from components containing Otto fuel. Safety, handling, and disposal of the Otto fuel shall be in accordance with NavWeps OP 3366. The Otto fuel shall not be reused for testing pumps or for any other purpose. The pump shall deliver the Otto fuel at the pressures specified in Table VI, within the respective parameters listed, when driven at 3880 ± 10 rpm, with bias pressure controlled within the respective parameters listed, and using the flow-limiting orifice specified in 4,4.3.1 d. During the test, the pump shall remain loaded, the orifice shall not be by-passed. Pump inlet pressure shall be maintained between 4 and 10 psig.

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TABLE VI

PREPRODUCTION TEST
OPERATING CHARACTERISTICS

Sequence	Bias Pressure Psig	Output Pressure Psig	Minimum Oper. Time At Each Sequence*
1	0	2345 ± 100	35 Sec.
2	85 ± 10	2475 ± 75	35 Sec.
3	225 ± 10	2675 ± 75	35 Sec.
4	445 ± 10	2925 ± 75	35 Sec.
5	665 ± 10	3155 ± 100	35 Sec.
6	1100 ± 10	3575 Min.	35 Sec.
7	665 ± 10	3155 ± 100	70 Sec.
8	445 ± 10	2925 ± 75	35 Sec.
9	225 ± 10	2675 ± 75	35 Sec.
10	85 ± 10	2475 ± 75	35 Sec.
11	0	2345 ± 100	35 Sec.

* Total operating time: 420 +30 -0 seconds. Time includes adjustment of bias pressure.

After each test, post-test servicing and maintenance shall be conducted in accordance with 4.5.7.

4.4.4.2 Vibration ready condition, The pump, in the ready condition, shall be mounted in a manner equivalent to that shown in Figure 1. The resonant modes in each of the 3 axes shall be determined by varying the frequency of applied vibration slowly from 5 cps to 500 cps. The pump shall ther be subjected to resonance and cycling vibration in each axis, in conformance with the times shown in Table VII. If more than 4 resonances are encountered on any axis, the 4 most severe resonances shall be chosen for the test. The resonant and cycling period, for each axis, shall be divided into 3 equal parts, which are conducted at -45° F ± 5° F, room temperature and +140° F ± 5° F respectively. For cycling, a logarithmic frequency sweep race shall be used if available; otherwise, other cycling rates of frequency change may be used. The sweep rate shall be such that 15 minutes actual vibration time is required to proceed from the lower to higher and back to the lower extremity of frequency range in accordance with the accelerations or double amplitudes defined in Table VIII.

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TABLE VII
VIBRATION TEST SCHEDULE

Number of Resonances	0	1	2	3	4
Total Vibration Time at Resonance *	-	30 min	1 hr	1-1/2 hr	2 hr
Cycling Time	3 hr	2-1/2 hr	2 hr	1-1/2 hr	1 hr

* 30 minutes each resonance

TABLE VIII
VIBRATION SCHEDULE

Frequency (cps)	5-22	22-40	40-74	74-500
Double Amplitude (in. ± 10%) Minimum	0.2		0.036	
Acceleration (g's ± 10%) Minimum		3		10

4.4.4.3 Acceleration, ready condition. The pump, in the ready condition, shall be subjected to the accelerations defined in Table IX after being secured to an acceleration equipment (see Figure 1). Acceleration shall be measured at the approximate geometric center of the pump.

TABLE IX
ACCELERATION SCHEDULE

Magnitude (g's ± 10%)	Acceleration Direction (See Figure 1)	Time
25	Longitudinal axis, Fwd	1 minute
35	Longitudinal axis, Aft	1 minute

4.4.4.4 Shock, ready condition. The pump, in the ready condition, shall be attached to a shock test machine (see Figure 1). Shock magnitudes and durations shall be measured by an accelerometer mounted on the mounting fixture. Tests on the pump shall be performed in accordance with Table X in the sequence shown.

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TABLE X

SHOCK TEST SCHEDULE

Test Sequence	Peak Magnitude (g's ± 10%)	Duration (ms)	Impact Axis	Direction Of Deceleration Force	Number Of Impacts
1	300	0.5 to 1.5	Longitudinal	Aft	3
2	85	0.5 to 1.5	Transverse A	Optional	3
3	85	0.5 to 1.5	Transverse B	Optional	3
4	45	18.0 to 20.0	Longitudinal	Fwd	3
5	60	8.0 to 10.0	Longitudinal	Aft	3
6	60	8.0 to 10.0	Longitudinal	Fwd	3
7	60	8.0 to 10.0	Transverse A	Optional	3
8	60	8.0 to 10.0	Transverse B	Optional	3

4.4,4.5 Temperature and humidity non-operating condition. Two pre-production pumps shall be subjected, in the non-operating condition, to temperature and humidity cycling in the following manner: The specimen shall be placed in a temperature chamber which has been stabilized at an environmental temperature of -45° F +0° F -5° F. The specimen shall remain in the environment for 6 hours. (As an alternative procedure, the specimen may be placed in the chamber, with the chamber at room ambient tempera-ture. Chamber temperature shall then be reduced to -45° F +0° F -5″ F within 2 hours maximum. The specimen shall thereafter remain in the chamber for 4 hours.) The specimen shall then be subjected to an environ-mental temperature of +140° F +5° F -0° F. Maximum transition time from the low temperature to the high temperature shall be 2 hours. The specimen shall remain in the high temperature for 4 hours. This cycle shall be performed 3 times; maximum transition time shall be 2 hours. The specimen shall subsequently be subjected to an environment of +110° F +5° -0° F at a minimum relative humidity of 9.5 percent for a minimum of 8 hours. There-after, one pump shall be operated in accordance with 4.4.4.6 and one pump shall be operated in accordance with 4.4.4.7. Prior to operation the pump may be wiped dry. Drying by an air blast or oven is not permitted.

4.4.4.6 Low temperature operating. Before testing, the pump shall be conditioned at a temperature of 0° F ± 5° F for a period of 1 hour. Immediately thereafter, the pump shall be operated in an environmental temperature of 0° F ± 5° F, for a period of 7 minutes. Otto fuel shall be supplied to the pump at 0° F ± 5° F. The pump shall be operated in accordance with and meet the requirements of 4.4.4.1. Post-test servicing and maintenance shall be performed in accordance with 4.5.7.

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4.4.4.7 High temperature operating. Before testing, the pump shall be conditioned at a temperature of +140° F ± 5° F for a period of 1 hour. Immediately thereafter, the pump shall be operated in an environmental temperature of +140° F ± 5° F for a period of 7 minutes. Otto fuel shall be supplied to the fuel pump at +140° F ± 5° F. The pump shall be operated in accordance with and meet the requirements of 4.4.4.1. Post-test servicing and maintenance shall be performed in accordance with 4.5.7.

4.4.4.8 Vibration, packed and packaged. The pump, packed and packaged in accordance with Section 5, shall be vibrated in such a manner as to determine the frequency producing maximum transmissibility within the frequency range of 5 to 60 cps. The container shall be fastened, in suitable fashion, to the table of a vibration testing device producing an essentially sine wave motion. The maximum permissible motion freedom shall be approximately 1/32 inch. The specimen shall then be vibrated for 1 hour at the frequency producing maximum transmissibility and at an input acceleration of 1 go This test shall be applied to each of the three principal orthogonal axes.

4.4.4.9 Shocks, Packed and Packaged. The pump, packed and packaged in accordance with Section 5, shall be subjected to free fall drop tests. The pump in its container shall be dropped from a height of 36 inches. A total of four edgewise drops shall be made. Two drops, each, shall be made on the bottom and top edges at points 180 degrees apart. In addition, flat drops shall be made, one each, on the bottom and on the side. The drops shall be made on an impact surface of essentially non-fielding material such as stone or concrete. The complete series of drops shall be performed under each of the following conditions: roan temperature, within 10 minutes after the container has been stabilized at -45° F ± 5° F, and within 10 minutes after the container has been stabilized at ±140° F = 5° F.

4.4.4.10 Operational service life. Each preproduction sample shall complete the life test requirement by being subjected to a total of ten 7-minute operational tests in accordance with 4.4.4.1 during the preproduction test schedule, Table V. Working fluid for these tests shall be Otto Fuel, Composition II. The number of additional operational tests needed to meet the life requirement is listed in Table V under “Operational Service Life”.

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4.4.5 Test data sheets. All test data shall be recorded on a data sheet by the supplier and maintained as part of the inspection records. In addition to the data required by 4.5.5 the data sheets shall contain, but are not limited to, the following information:

a.	Low temperature operating test:	Pump conditioned for: _____hrs. at _____° F Fuel temperature _____° F
b.	High temperature operating test:	Pump conditioned for: _____hrs. at _____° F Fuel temperature _____°F

4.4.6 Determination of defect or failure. Failure of a pump to successfully pass any of the tests in 4.4 through 4.4.4.10 shall be cause for rejection of the preproduction sample.

4.5 Acceptance inspection. Each pump produced under the contract or purchase order shall be subjected to the following tests and examinations.

4.5.1 Tests and examinations.

a. Operational test
b. Visual examination

4.5.2 Test conditions. Unless otherwise specified, the pump shall be subjected to the acceptance tests under the following conditions:

a.	Temperature:	Room ambient +32° F to +98° F
b.	Altitude:	Normal ground atmospheric pressure
c.	Vibration:	None
d.	Humidity:	Room ambient

4.5.3 Test equipment and set-up. The test equipment set-up shall be in accordance with Figure 3.

a. Bias pressure control, regulated inert gas supply: Adjustable output range, 0 to 1250 psig

b. Low pressure inert gas supply: Adjustable range, 0 *to* 25 psig nominal

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c. Gages:

Item	Range Psig	Accuracy Full Scale	Size Dia.
A	0-1250	± 1%	6 in.
B	0-5000	± 1/2%	6 in.
C	0-25	± 5%	4 in.

d. Relief valve: Cracking pressure, 4500 ± 100 psig

e. Accumulator: Size, 10 cubic inches; test pressure 8000 psig minimum

f. Working fluid supply: Capable of supplying to the pump a 5 percent solution, by weight, of water-soluble oil in water at a rate of 2 gpm at 7 ± 3 psig and at any temperature between +32° and +98° F

g. Drive unit: 5 horsepower synchronous electric, nominal

h. Outlet filter: 25 micron maximum, 5 square inches of screen area minimum, 4000 psig minimum working pressure

i. Orifice, flow-limiting: The fixed orifice, when connected in the test set-up, Figure 3, shall produce a flow rate as defined by:

$P = KQ^2 \quad P > 2500$ psig

P = pressure-psig

Q = flow, gallons per minute working fluid

K = 1500 ± 94

j. Inlet filter: 50 to 80 micron

4. 5.4 Test procedures.

4.5.4.1 Operational test. The pump, containing twenty cubic centimeters of oil, see 6.7, shall be mounted in the test stand and connected to the auxiliary equipment. Prior to testing, the flow rate shall be preset as folows: remove the Adjusting Screw Nut (Figure 1), apply a pressure Of 300 ± 10 psig to the bias port. With the pump being driven at 3880 ± 10 rpm, set the inlet pressure to 7 ± 3 psig. Position the Valve Adjusting Screw (Figure I) to obtain an outlet pressure of 2775 ± 10 psig. Replace the Adjusting Screw Nut and secure with a lockwire. The pump shall be operated in accordance with the sequence indicated in Table XI. The pump shall deliver the working fluid at the pressures specified in Table XI, when the pump is driven at 3880 ± 10 rpm, bias pressure is controlled within the respective parameters and the inlet pressure in maintained between 4 and 10 psig. The orifice shall not be by-passed thus unloading the pump during the test.

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TABLE XI

OPERATING CHARACTERISTICS

Sequence	Bias Pressure Psig	Output Pressure Psig	Minimum Operating Time At Each Sequence
1	0	2345 ± 100	10 sec.
2	85 ± 10	2475 ± 75	10 sec.
3	225 ± 10	2675 ± 75	10 sec.
4	445 ± 10	2925 ± 75	10 sec.
5	665 ± 10	3155 ± 100	10 sec.
6	1100 ± 20	3575 Min.	10 sec.
7	665 ± 10	3155 ± 100	15 sec.
8	445 ± 10	2925 ± 75	10 sec.
9	225 ± 10	2675 ± 75	10 sec.
10	85 ± 10	2475 ± 75	10 sec.
11	0	2345 ± 100	10 sec.

4.5.4.2 Visual examination. There shall be no evidence of working fluid leakage from the pump during the acceptance test.

4.5.5 Test data sheets. All test data shall be recorded on data sheets by the supplier. Data sheets shall contain, but are not limited to, the following information:

a. Date _____
b. Date of calibration; Instrumentation _____ Orifice____,
c. Preset adjustment:
Bias pressure _____ psig, Output pressure _____ psig .
d. Test Data

Sequence	Bias Pressure Psig	Output Pressure Psig
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
11		

e. Working fluid leakage _____
f. Post-test servicing _____

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4.5.6 Determination of defect or failure. Failure of a pump to successfully pass the tests of 4.5.4.1 or 4.5.4.2 shall be cause for rejection of that pump.

4.5.7 Post-test servicing and maintenance. The pump shall be disassembled, cleaned, and re-assembled after being tested. The corrosion preventive solvent used shall be in accordance with 6.5. All exposed O-rings shall be replaced. Nut, Adjusting Screw Dug. 2526741, Figure 1, shall not be removed.

4.6 Periodic Production inspection. Periodic production samples shall be selected in accordance with 4.3.2. The samples shall be tested in accordance with the following plan:

a. The Sample of 1 unit from the first group of 30 units will be called the first Sample and will undergo the tests of Table XII for Sample 1. The sequence of tests will be as shown in Table XII.

b. The Sample of 1 unit from the second and third groups of 30 units will be called the second and third Samples and will undergo the tests of Table XII for Samples 2 and 3 respectively. The sequence of tests will be as shown in Table XII.

c. The next Sample will be selected from the next group of units regardless of whether the shift to Step 2 was made. This Sample will be called the fourth Sample and will undergo the tests of Table XII for Sample 4, etc.

TABLE XII

PERIODIC TEST SEQUENCE

Tests	Sample and Test Sequence 1, 4, Etc.	2, 5, Etc.	3, 6, Etc.
Vibration, Ready Condition		1*	1*
Acceleration, Ready Condition			2*
Shock, Ready Condition	1*		3*
Temperature and Humidity		2	4
High Temperature Operating	2		5
Low Temperature Operating		3	6
Operational Test	3	4	
Operational Service Life			7

* Visual Check After Testing

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4.6.1 Test conditions. Unless otherwise specified, the pump shall be subjected to the periodic tests under the following conditions:

a.	Temperature:	Room ambient +32° F to +98° F
b.	Altitude:	Normal ground atmospheric pressure
c.	Vibration:	None
d.	Humidity:	Room ambient

4.6.2 Test equipment and set-up. The test equipment and set-up for the periodic tests shall be in accordance with 4.4.3.

4.6.3 Test procedures.

4.6.3.1 Vibration, ready condition. Vibration testing of the pump, in the ready condition, shall be performed in accordance with 4.4.4.2.

4.6.3.2 Acceleretion, ready condition. Acceleration testing of the pump, in the ready condition, shall be performed in accordance with 4.4.4.3.

4.6.3.3 Shock, ready condition. Shock testing of the pump, in the ready condition, shall be performed in accordance with 4.4.4.4.

4.6.3.4 Temperature and humidity, non-operating condition. Temperature and humidity testing of the pump, in the non-operating condition, shall be performed in accordance with 4.4.4.5.

4.6.3.5 High temperature operating. The pump shall be subjected to the high temperature operating test in accordance with 4.4.4.7.

4.6.3.6 Low temperature operating. The pump shall be subjected to the low temperature operating test in accordance with 4.4.4.6.

4.6.3.7 Operational test. The pump shall be subjected *to* the operational test in accordance with 4.4.4.1.

4.6.3.8 Operational service life. The pump shall be subjected to 6 additional operational tests in accordance with 4.4.4.1.

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4.6.4 Test data. Test data shall be recorded in accordance with 4.4.5, as appropriate, and maintained as a part of the inspection record.

4.6.5 Determination of defect or failure. Failure of the pump to successfully pass the tests in 4.6.3.1 through 4.6.3.8 shall be cause for withholding acceptance of all units on hand. Corrective action shall be taken in accordance with 4.3.2.

4.7 Examination of first echelon repair parts.

4.7.1 General.

4.7.1.1 The information presented in this section is applicable to all components assembled into the end product and to first echelon repair parts. The information is divided into the following major divisions.

a. Parts covered by separate CD'S
b. Parts which are manufactured in accordance with Federal or Military specifications which contain adequate quality assurance provisions.
c. Minor hardware items

4.7.1.2 Unless otherwise specified, sampling plans and procedures set forth in MIL-STD-105, Inspection Level II, shall be used in determination of acceptability of the first echelon repair parts and all components assembled into the end product. An AQL of 1.0 shall be used for major items of inspection, an AQL of 2.5 shall be used for minor items of inspection.

4.7.1.3 The verification of the quality of repair parts is mandatory and shall be performed whether the item is a part Of a repair kit or is to be assembled into the end product.

4.7.2 Separate Classification of Defects (CD's). The components and assemblies referenced within this subsection shall be inspected in accordance with the applicable CD's. For inspection of repair parts use:

PART	BU WEPS CD
Spring, Ball Guide	1556991
Spring, Piston	1557047
Bearing, Needle	1557048
Packing, Preformed	1557052
Valve Assembly	1557063

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PART	BU WEPS CD
Shaft, Pump	1557065
Body, Fuel Pump	1568879
Block, Piston	1568892
Head Pump	1568924
Pump Assembly, Fuel	1568925
Seal, Lower	2526584
Piston	2526593
Cam	2526597
Screw, Cap, Soc Hd, Hex	2526611
Retainer, Seal	2526612
Plate, Thrust	2526613
Bearing, Needle	2525614
Bearing, Thrust	2526616
Washer, Thrust	2526617
Seal, Plain Encased	2526624
Packing, Preformed	2526718
Head Assembly	2526724
Fir., Valve	2526738
Guide, Ball	2526739
Screw, Valve Adj.	2526740
Nut, Adj. Screw	2526741
Ball	2526742
Seal, Upper	2526834
Key, Woodruff	2571159

4.7.3 Components covered by Federal and Military Specifications. The components and parts referenced within this subsection shall be inspected in accordance with the applicable specifications. For inspection of repair parts use:

PART	SPECIFICATION
Lubricant	MIL-G-4343
Oil	MIL-L-15016
Adapter	MIL-F-18806
Elbow	MIL-F-18866

4.7.4 Miscellaneous hardware. The items referenced within this subsection shall be inspected as to type and quantity.

PART	IDENTIFICATION
Packing, Preformed	ARP 568-008
Ring, Retaining	MS16626-1056
Lockwire	MS20995-C32
Packing, Preformed	MS29561-015,021
Fitting, Ball Design	MS35670-2

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PART	IDENTIFICATION
Set Screw, Cup Point	MS51018-49
Packing, Preformed	NAS617-3,4,6
Washer, Flat	NAS620C10L
Screw,Cap, Soc Hd, Hex	NAS1351-3,16

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified. See 6.2. Unit package shall consist of one pump.

5.1.1 Level A. Preservation and packaging of equipment for level A shipment shall be in accordance with MIL-P-116. Cleaning shall be accomplished in accordance with Process C-1 and drying, if required, shall be accomplished with Method D-1, or D-4. Preservation and packaging shall be Method IA.

5.1.2 Level B. Not applicable.

5.1.3 Level C. Cleaning, drying, presentation and packaging shall be accomplished in accordance with the supplier's own commercial practice when such practices have been approved by the procuring activity.

5.2 Packing. Packing shall be level A or C as specified. See 6.2. Shipping container shall contain one unit package.

5.2.1 Level A. The equipment, when preserved and packaged in accordance with 5.1.1, shall be overpacked in containers complying with MIL-P-116, Method II D.

5.2.2 Level B. Not applicable.

5.2.3 Level C. The equipment, when preserved and packaged in accordance with 5.1.3, shall be overpacked in compliance with Uniform Freight Classification Rules or other common carrier regulations applicable to the mode of transportation.

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5.2.4 Cushioning or dunnage. Supplier shall provide sufficient cushioning or dunnage material to adequately protect the pump from damage due to shock and vibration loads encountered during all modes of handling and transportation. Shock and vibration tests shall be in accordance with this specification.

5.3 Marking. In addition to the markings required by the contract or order, unit packages and shipping container shall be marked in accordance with requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. The fuel pump is intended for use in a torpedo propulsion system to pump a liquid monopropellant fuel, Otto Type II, WS-2106.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification
b. Assigned activity for preproduction tests
c. Any exceptions to this specification
d. Inspection data required
e. Required level of preservation, packaging and packing

6.3 Definitions.

6.3.1 Ready. The condition of the pump when fully assembled after completion of servicing and maintenance.

6.3.2 Operating. The condition of the pump when operating.

6.3.3 Non-operating. The condition of the pump where stored, transported, or handled. Tests on the pump, in the non-operating condition, shall be performed without packaging or packing.

6.3.4 Vibration. Vibration is a sinusoidal motion defined by amplitude and frequency. The amplitude may refer to displacement or to acceleration. The displacement is given in inches peak-to-peak or the corresponding maximum acceleration in gravitational units. The frequency is given in cycles per second.

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6.3.5 Acceleration (steady state). Acceleration (steady state) is the time rate of change of velocity of a body resulting from the application of a sustained external force with a uniform build-up to peak in more than 0.010 seconds. Acceleration is given in gravitational units.

6.3.6 Shock. Shock is an abrupt change in the motion of a body resulting from the sudden application of a large external force of short duration. Shock, in general, can be described as a build-up to 90 percent of peak force in less than 0.010 seconds. Shock is described as a pulse in units of gravitational acceleration, duration in milliseconds and shock wave form equivalent to one half sine wave, Duration is defined as the interval between the point of zero amplitude, from which the shock waveform builds up, to its decay to zero amplitude.

6.3.7 Temperature. The temperature of the environment to which the pump is exposed, is expressed in degrees Fahrenheit.

6.3.8 Humidity. Humidity is the relative humidity of the environment to which the pump is exposed, expressed in percent.

6.4 Weight. The weight of the pump shall be 3.62 ± 0.20 pounds.

6.5 Corrosion preventive solvent. The corrosion preventive solvent shall consist of a solution of 4 parts mineral spirits and 1 part Steelgard #1301.

6.6 Storage life.

6.6.1 Packed and packaged condition. The pump shall be capable of withstanding storage for 5 years in the packed and packaged condition and, thereafter, meet the performance requirements of this specification.

6.7 Lubricating oil. The lubricating oil, in addition to meeting the requirements of MIL-L-15016-3065, shall have a minimum viscosity index of 100.

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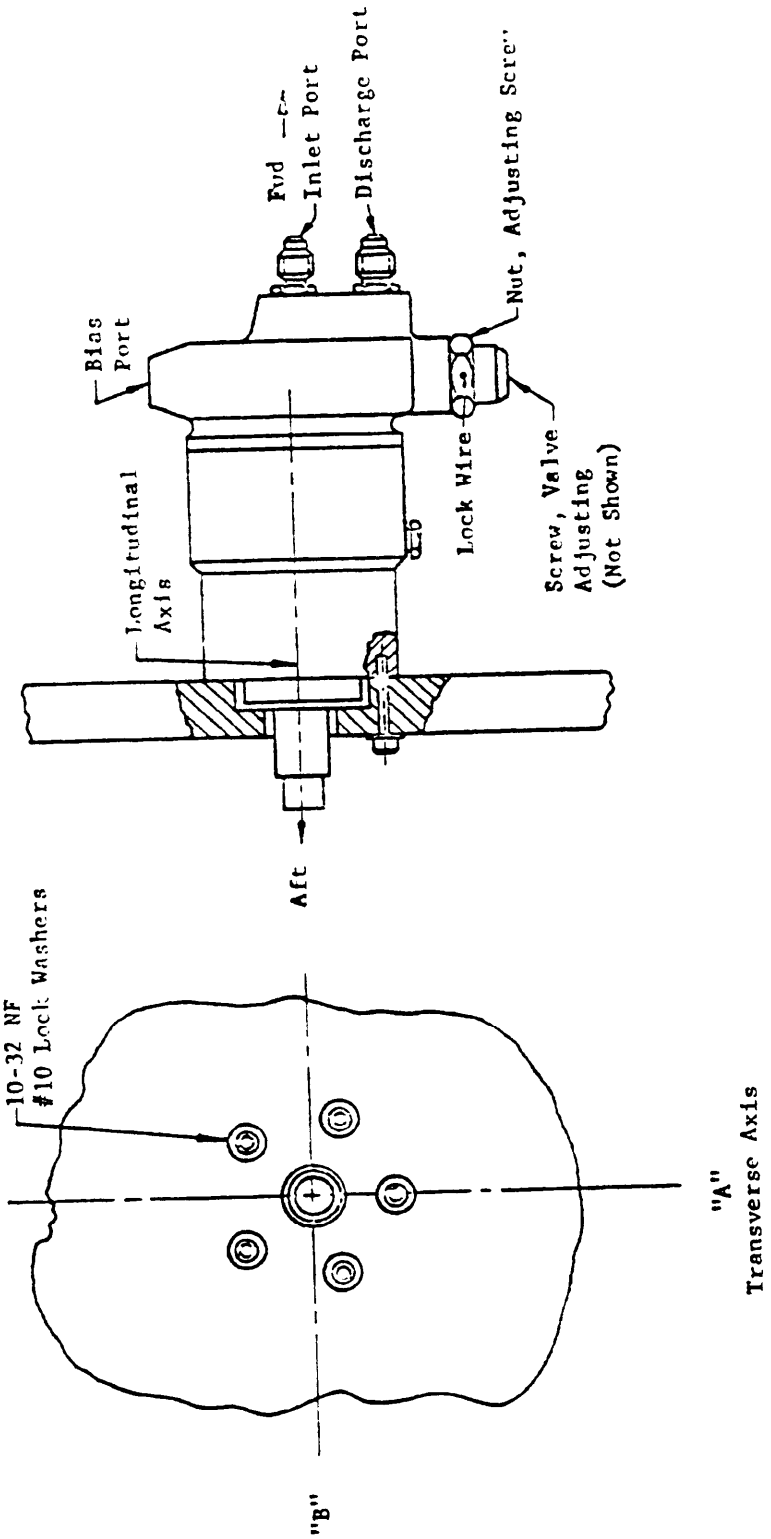


Figure 1, Support Frame

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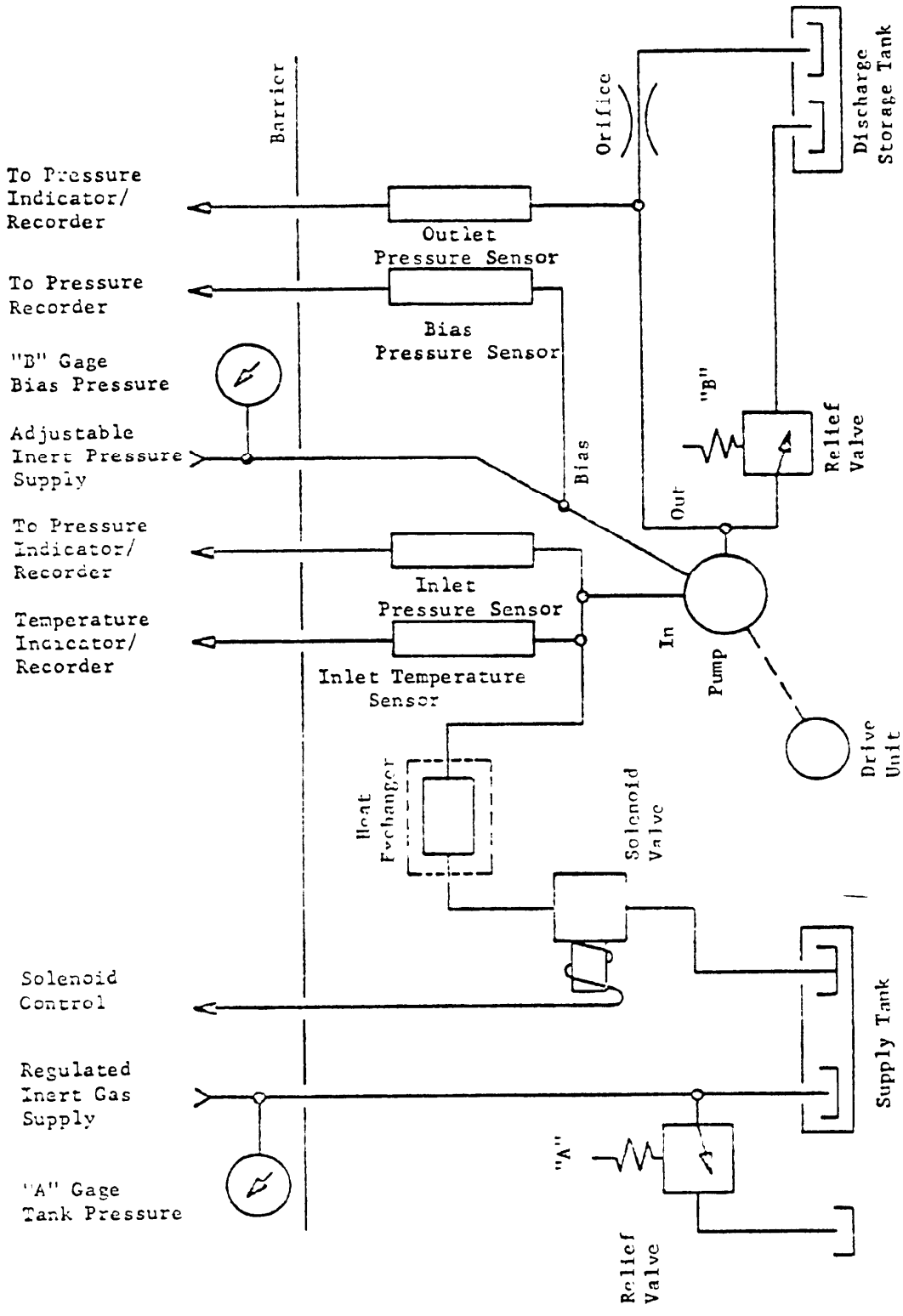


Figure 2, Test Set-Up for Preproduction Inspection

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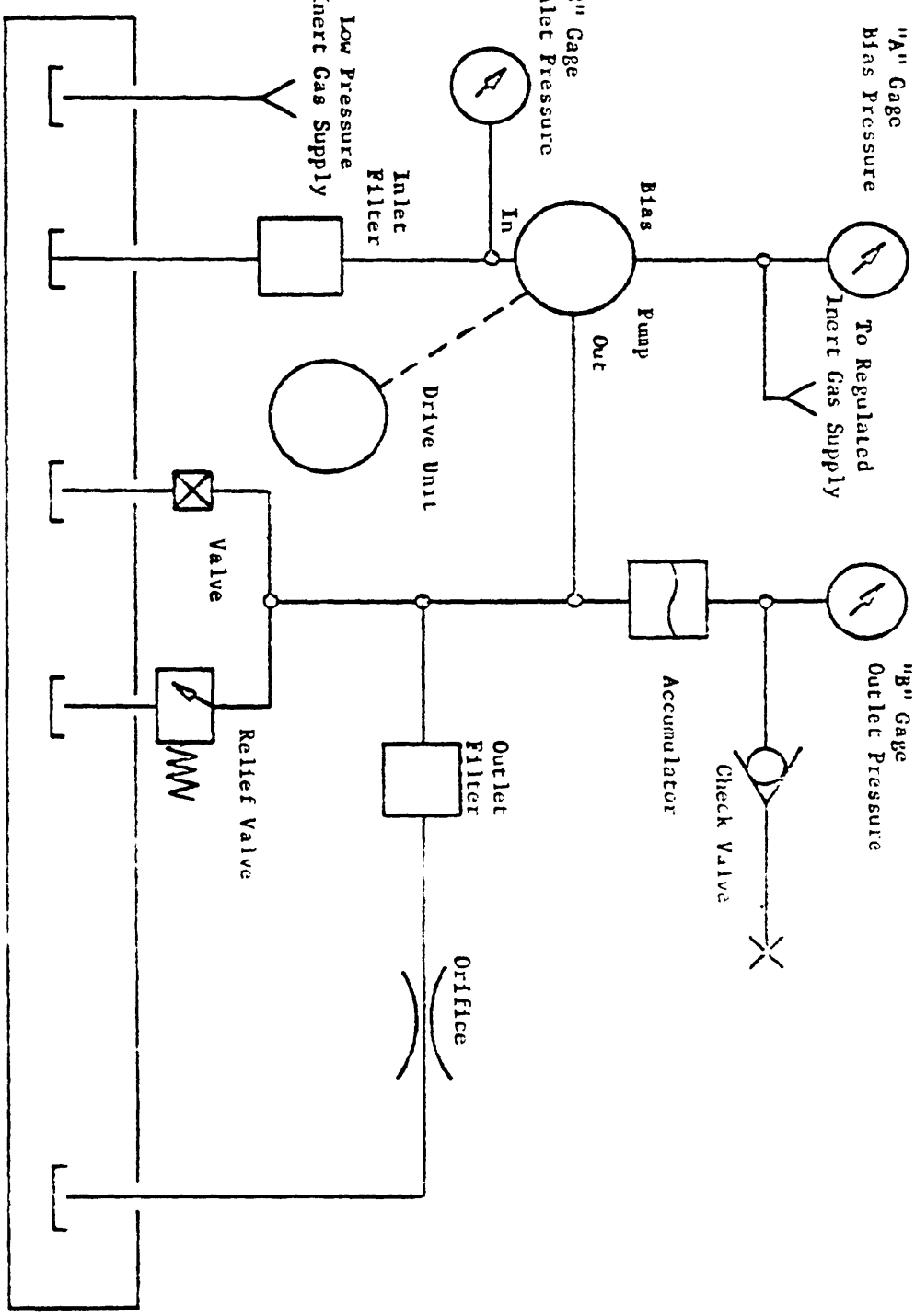


Figure. 3, Test Set-Up for Acceptance Inspection

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-P-82436(OS) Pump, Fuel

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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